



Vlad-Emil Flore

Backend Software Engineer (Python, Java, GenAI)

DETAILS

✉ flore.vlad@gmail.com

📍 Barcelona

🌐 Romania

EDUCATION

Machine Learning Specialization

DeepLearning.AI & Stanford
Online
2026

Master of Science in Informatics

West University of
Timișoara
2008 – 2010

Focus on Artificial Intelligence
and Distributed Computing

Grade: 9/10

Bachelor of Science in Informatics

West University of
Timișoara
2004 – 2008

Grade: 9.67/10

PROFILE

- Backend Software Engineer specializing in **Python, Java**, and **GenAI-powered** solutions, building scalable, cloud-native applications and modernizing complex legacy systems
- Versatile project experience across greenfield and brownfield initiatives in public sector services, content management systems, and intelligence document processing solutions, including **AI-assisted** data extraction, classification, and automation
- Proven leader in agile development environments, driving team collaboration, ensuring code quality, and consistently delivering projects on schedule

LANGUAGES

Romanian	NATIVE
English	C1
German	B2+
Spanish	A2

SKILLS

-  Programming Languages (Python, Java)
-  Machine Learning and Generative AI
-  Testing (JUnit, Mockito, AssertJ, Pytest, Testcontainers)
-  Coding Practices (Clean Code, TDD, SOLID)
-  Web API (RESTful APIs, GraphQL, OpenAPI, Swagger, PostGraphile)
-  Web Frameworks (Spring Framework, Spring Boot, Django, FastAPI)

EMPLOYMENT HISTORY

INDEPENDENT SOFTWARE ENGINEER

Freelancing, Remote (Barcelona)
January 2025 - present

- Developed and maintained **scalable Web APIs** in Python and Java, including integrations with external platforms and services
- Designed and built an **AI medical coding** platform that turns free-text diagnoses into standard medical codes using **LLM- and RAG-powered** search, and contributed to a **GenAI document understanding** platform that extracts and classifies information from documents
- Implemented **human-in-the-loop** review, letting experts check, correct, and confirm low-confidence AI results, with corrections feeding back to improve future predictions
- Delivered both platforms as secure, containerized services with pluggable components, automated releases, and thorough documentation
- Worked as a **Java Domain Expert** through Mercor, contributing to AI training, code evaluation, and technical assessment workflows for LLM systems
- Led documentation modernization efforts, including migration to modern documentation platforms to improve maintainability, structure, and multi-version documentation support
- Created software engineering **educational content** on Python, Java, Data Structures & Algorithms, and Git, including animated video series, interactive quizzes, and flashcards

TECH STACK: Python, FastAPI, Django, Java, Spring (Framework, Boot, AI), PyScript, Manim, OpenAPI, Generative AI (LLM, RAG, Agentic Systems), LangChain(4j), LangGraph, LangSmith, Ollama, Weaviate, PostgreSQL (with pgvector), MySQL, Azure Cosmos DB, SQLAlchemy, Alembic, Azure Blob Storage, OAuth, OIDC, JWT, OpenTelemetry, Docker (Compose), Kubernetes, GitHub Actions, AWS (EC2, S3, IAM, Lambda, ECR, ECS, EKS), Antora, Docusaurus

 Messaging and Event-Sourcing Systems
(RabbitMQ, EventStoreDB)

 Databases & ORM
(PostgreSQL, Oracle, (Spring Data) JPA/Hibernate, SQLAlchemy)

 Cloud (AWS, Kubernetes, Keda, Docker)

 Build Tools (Maven, Gradle, Bazel)

 CI/CD (Git, GitHub, GitHub Actions, GitLab, TeamCity, Argo CD, Jenkins, Helm, Ansible)

 Team Leadership, Mentoring & Knowledge Sharing

LINKS

 Certifications

 LinkedIn

 GitHub

SOFTWARE ENGINEER

Employment, ExB Group, Remote (Nürnberg, Barcelona)

March 2021 – November 2024

- Implemented key features in Python and Java within a microservices-based event-driven architecture for a web-based Intelligent Document Processing application
- Designed and developed scalable, decoupled microservices in Python and Java, enhancing system robustness and performance (documents upload, archiving and download, image processing, REST and GraphQL APIs, notifications, AI models deployment)
- Deployed services in a cloud-native environment using Docker and Kubernetes, ensuring high availability and scalability
- Led internal technical workshops focused on clean code principles and improving overall code quality
- Conducted regular code reviews and participated in pair programming to foster collaboration and knowledge sharing
- Advocated for tooling and workflow improvements to streamline the developer experience and reduce friction in daily tasks

TECH STACK: Python, Java, Django, Spring Framework, Spring Boot, Akka, GraphQL, PostGraphile, PostgreSQL, EventStoreDB, RabbitMQ, Docker, Kubernetes, Helm, Keda, AWS S3, MinIO, AWS ECR, AWS ECS, AWS EKS, Git, GitLab, TeamCity, Argo CD, Grafana, AsciiDoc, Antora, Linux, JUnit

EDUCATIONAL CONTENT CREATOR

Contract, Hyperskill, Remote (Barcelona)

October 2023 – March 2024

- Created and enhanced educational content for the Spring Boot track offered by JetBrains Academy through the Hyperskill platform, contributing to high-quality, hands-on learning experiences for aspiring developers

TECH STACK: Java, Spring Framework, Spring Boot

LEAD SOFTWARE DEVELOPER

Contract, IT-Systemhaus der Bundesagentur für Arbeit, on-site (Nürnberg)

September 2019 – April 2020

- Served as Lead Developer for one of two agile development teams, guiding technical direction and team collaboration
- Defined and evolved the application's architecture to support scalability and maintainability
- Promoted clean code and consistent development practices to maintain code quality
- Collaboratively decomposed features into manageable tasks, ensuring effective sprint planning and delivery
- Implemented and tested application features across the full stack, maintaining close alignment with business requirements

TECH STACK: Java SE 8, Java EE 7, Angular, TypeScript, JavaScript, JUnit, Cucumber, Jenkins, Docker, Linux, Oracle Database, Git

FULL STACK DEVELOPER

Contract, Bundesamt für Migration und Flüchtlinge (BAMF), on-site (Nürnberg)

May 2018 – September 2019

- Implemented features across both frontend and backend of a web application
- Improved and maintained project documentation to support developer onboarding and knowledge sharing
- Managed DevOps-related tasks and supported deployment activities to ensure smooth delivery and system stability

TECH STACK: Java SE 8, Spring Framework, Spring Boot, JUnit, Angular, TypeScript, JavaScript, Ansible, Jenkins, Docker, Oracle Database, Hibernate, Linux, Git

SOFTWARE DEVELOPER

Contract, IT-Systemhaus der Bundesagentur für Arbeit, on-site (Nürnberg)

May 2014 – February 2018

- Developed a publishing workflow for web content within Oracle CMS, streamlining the content creation and approval process
- Extended the CMS by implementing custom tools and features to support efficient content management and improve the user experience for content editors

TECH STACK: Oracle WebCenter Content, Java, HTML, CSS, JavaScript, Oracle Database, Linux, CVS, Git

SENIOR TECHNICAL CONSULTANT

Employment, virtual7 GmbH, on-site (Nürnberg)

February 2019 – May 2020

- Delivered software solutions across various client projects as part of agile development teams
- Implemented new features, resolved bugs, and participated in the full SDLC
- Acted as a software consultant, adapting quickly to different domains and tech stacks

SOFTWARE DEVELOPER / SENIOR TECHNICAL CONSULTANT

Employment, virtual7 Romania, on-site (Timișoara)

February 2008 – January 2019

- Delivered software solutions across various client projects as part of agile development teams
- Implemented new features, resolved bugs, and participated in the full SDLC across various client engagements
- Acted as a software consultant, adapting quickly to different domains and tech stacks